

Problema 1

Fie $(\pi_k)_{k \in \mathbb{N}}$ polinoamele ortogonale Legendre monice.

(a) Aratati ca polinoamele

$$\pi_k^+(t^2) = \pi_{2k}(t)$$

sunt ortogonale monice pe $[0, 1]$ in raport cu ponderea $w(t) = \frac{1}{\sqrt{t}}$

Solutie:

Stim ca:

$$\int_{-1}^1 \pi_{2k}(t) \pi_{2j}(t) dt = 0 \text{ pentru } k \neq j \text{ (conditia de ortogonalitate a } \pi_{2k} \text{ pe } [-1, 1]) \quad (1)$$

Descompunem integrala in 2 subintervale: $[-1, 0], [0, 1]$. Avand in vedere ca toate puterile pare ale lui t sunt functii simetrice (pare), rezulta ca $\pi_{2k}(-t) = \pi_{2k}(t)$, deci:

$$\int_{-1}^1 \pi_{2k}(t) \pi_{2j}(t) dt = 2 \int_0^1 \pi_k^+(t^2) \pi_j^+(t^2) dt \quad (2)$$

Vom face schimbarea de variabila $u = t^2$, $dt = \frac{du}{2\sqrt{u}}$

$$\int_{-1}^1 \pi_{2k}(t) \pi_{2j}(t) dt \stackrel{(2)}{=} 2 \int_0^1 \pi_k^+(t^2) \pi_j^+(t^2) dt = 2 \int_0^1 \frac{\pi_k^+(u) \pi_j^+(u)}{2\sqrt{u}} du = \int_0^1 \pi_k^+(u) \pi_j^+(u) \cdot \frac{1}{\sqrt{u}} du \quad (3)$$

Din (1), (3) $\Rightarrow$ polinoamele π_k^+ sunt ortogonale monice (monicitate mostenita direct de la $\pi_{2k}(t)$) pe $[0, 1]$ in raport cu ponderea $w(t) = \frac{1}{\sqrt{t}}$

(b) Stabiliti formula de cuadratura

$$\int_0^1 \frac{1}{\sqrt{x}} f(x) dx = 2 \sum_{k=1}^n A_k f(t_k^2) + R_n(f)$$

unde A_k si t_k , $k = 1, \dots, 2n$ sunt coeficientii si respectiv nodurile formulei de cuadratura

Gauss-Legendre cu $2n$ noduri

Solutie:

Vom face schimbarea de variabila $x = t^2$, $dx = 2t dt$

$$\int_0^1 \frac{f(x)}{\sqrt{x}} dx = \int_0^1 \frac{f(t^2)}{t} \cdot 2t dt = 2 \int_0^1 f(t^2) dt.$$

Extindem integrala pe intervalul $[-1, 1]$ folosind simetria (deoarece $f(t^2)$ este functie para):

$$2 \int_0^1 f(t^2) dt = \int_{-1}^1 f(t^2) dt.$$

Aplicand acum formula Gauss-Legendre cu $2n$ noduri, rezulta

$$\int_0^1 \frac{1}{\sqrt{x}} f(x) dt = \int_{-1}^1 f(t^2) dt = \sum_{k=1}^{2n} A_k f(t_k^2) + R_{2n}(f).$$

Avand in vedere simetria formulei Gauss-Hermite, avem $t_k = -t_{n-k}$ si $A_k = A_{n-k}$, $k = 1, \dots, 2n$ rezulta $t_k^2 = t_{n-k}^2$ si

$$\sum_{k=1}^{2n} A_k f(t_k^2) = 2 \sum_{k=1}^n A_k f(t_k^2).$$

Pentru rest, daca $f \in C^{4n}[-1, 1]$ exista $\xi \in (-1, 1)$ astfel incat

$$R_n(f) = R_{2n}(f(t^2)) = \frac{(f(t^2))^{(4n)}}{(4n)!} \int_{-1}^1 [\pi_{2n}(t)]^2 dt = \frac{f^{(4n)}(\xi)}{(4n)!} \int_{-1}^1 [\pi_{2n}(t)]^2 dt$$

(c) Implementati (in MATLAB) o formula de cuadratura de tip Gauss pentru integrala $\int_0^1 \frac{1}{\sqrt{x}} f(x) dx$ folosind ideea de la punctul (b)

Solutie:

Folosim Gauss-Legendre. Noile noduri vor fi t_k^2 , iar noi coeficienti $2A_k$, $k = 1, \dots, n$

(vezi sursa Gauss_rsqr mai jos la Functii)

(d) Folosind functia de la punctul (c) calculati $\int_0^1 \frac{\sin(x)}{\sqrt{x}} dx$ cu 8 zecimale exacte.

```
%Calculeza \int_0^1 1/sqrt(x)f(x)dx
%cu ajutorul polinoamelor Legendre
format long
f=@(x) sin(x);
[gn,gc]=Gauss_rsqr(20);
vi=vquad(gn,gc,f);
[gn2,gc2]=Gauss_Jacobi_ab(20,0,-1/2,0,1);
vi2=vquad(gn2,gc2,f);
fprintf("Rezultat cu Legendre: %.8f\n", vi)
```

Rezultat cu Legendre: 0.62053660

```
fprintf("Rezultat cu Jacobi: %.8f\n", vi2)
```

Rezultat cu Jacobi: 0.62053660

Problema 2

Dorim sa calculam $\frac{1}{\sqrt{a}}$, pentru $a > 0$

(a) Pornind de la o ecuatie convenabila si folosind metoda lui Newton, deduceti o metoda pentru calculul lui $\frac{1}{\sqrt{a}}$

Solutie:

Alegem ecuati convenabila:

$$f(x) = \frac{1}{x^2} - a = 0 \Rightarrow \text{Solutia } x = \frac{1}{\sqrt{a}}$$

Aplicam metoda lui Newton:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, f'(x) = -\frac{2}{x^3}$$

$$x_{n+1} = x_n - \frac{\frac{1}{x_n^2} - a}{-\frac{2}{x_n^3}} = x_n + \frac{x_n}{2} (1 - ax_n^2)$$

$$\text{Formula finala: } x_{n+1} = \frac{x_n}{2} (3 - ax_n^2)$$

(b) Pentru ce valori ale lui x_0 metoda converge?

Solutie:

Scriem iteratia sub forma $x_{n+1} = F(x_n)$ cu $F(x) = \frac{x}{2} (3 - ax^2)$

Stim $x > 0$ (1)

$$\text{Din conditia de contractie } |F'(x)| < 1 \Rightarrow \left| \frac{3}{2} - \frac{3}{2}ax^2 \right| < 1 \Rightarrow \frac{1}{3} < ax^2 < \frac{5}{3} \quad (2)$$

$$(1), (2) \Rightarrow x_0 \in \left(\frac{\sqrt{3}}{3\sqrt{a}}, \frac{\sqrt{15}}{3\sqrt{a}} \right) \quad (3)$$

Impunand conditia $F(x) > 0 \Rightarrow \frac{x}{2}(3 - ax^2) > 0$ (4)

$$(1), (4) \Rightarrow 3 - ax^2 > 0 \Rightarrow x^2 < \frac{3}{a} \Rightarrow x < \sqrt{\frac{3}{a}} \quad (5)$$

Daca $0 < x_0 < \frac{1}{\sqrt{a}}$, atunci se respecta convergenta monotona

Pentru $x_0 \approx 0, F(x) = 3\frac{x}{2}$, ceea ce duce la solutii eronate

$$(3), (5) \Rightarrow \text{intervalul optim pentru } x_0 : x_0 \in \left(\frac{\sqrt{3}}{3\sqrt{a}}, \sqrt{\frac{3}{a}} \right)$$

(c) Folositi iteratia de la (a) pentru a da o metoda de calcul al radicalului fara impartiri.

Solutie:

Funcții:

```
function [gn, gc] = Gauss_rsqrtn(n)
%formula Gauss pentru w(t) = t ^(-1/2)
%foloseste polinoame Legendre
[nds, coeffs] = Gauss_Legendre(2*n);
gn = nds(1:n).^2;
gc = 2 * coeffs(1:n);
end
```

```
function [g_nodes,g_coeff]=Gauss_Legendre(n)
%GAUSS-LEGENDRE - Gauss-Legendre nodes and coefficients
%a
beta=[2,(4-(1:n-1).^(-2)).^(-1)];
alpha=zeros(n,1);
[g_nodes,g_coeff]=Gaussquad(alpha,beta);
end
```

```
function [g_nodes,g_coeff]=Gaussquad(alpha,beta)
%GAUSSQUAD - generate Gaussian quadrature formula
%computes nodes and coefficients for
```

```

%Gauss rules given alpha and beta
%method - Jacobi matrix
n=length(alpha); rb=sqrt(beta(2:n));
J=diag(alpha)+diag(rb,-1)+diag(rb,1);
[v,d]=eig(J);
g_nodes=diag(d);
g_coeff=beta(1)*v(1,:).^2;
end

```

```

function [g_nodes, g_coeff] = Gauss_Jacobi_ab(n, alpha, beta, a, b)
    [g_nodes_j, g_coeff_j] = Gauss_Jacobi(n, alpha, beta);
    g_nodes = (b-a)/2 * (g_nodes_j + 1) + a;
    g_coeff = g_coeff_j * ((b-a)/2)^(alpha+beta+1);
end

```

```

function [g_nodes,g_coeff]=Gauss_Jacobi(n,a,b)
    %Gauss-Jacobi - Gauss-Jacobi nodes and coefficients
    %weight function w(t)=(1-t)^a(1+t)^b
    if nargin<2, a=0; end; if nargin<3, b=a; end
    if((n<=0)|| (a<=-1)|| (b<=-1))
        error('parameter(s) out of range')
    end
    a0=(b-a)/(b+a+2);
    b0=2^(a+b+1)*gamma(a+1)*gamma(b+1)/gamma(a+b+2);
    if n==1
        alpha=a0; bet=b0;
    else
        k1=1:n-1;
        k2=2:n-1;
        b1=4*(1+a)*(1+b)/((2+a+b)^2*(3+a+b));
        if a==b
            alpha=zeros(1,n);
        else
            alpha=[a0,(b^2-a^2)./(2*k1+a+b)./(2*k1+a+b+2)];
        end
        bet=[b0,b1,4*k2.*(k2+a+b).*(k2+a).*...
            (k2+b)./(2*k2+a+b-1)./(2*k2+a+b).^2./(2*k2+a+b+1)];
    end
    [g_nodes,g_coeff]=Gaussquad(alpha,bet);
end

```

```

function I=vquad(g_nodes,g_coeff,f)
I=g_coeff*f(g_nodes);
end

```